Full-Horseshoe Algebraic Exhaustion at the Area-Preserving Hénon Parameter

Liang Wang
School of Artificial Intelligence and Automation
Huazhong University of Science and Technology
Wuhan 430074, P. R. China

August 14, 2026

Abstract

For the area-preserving quadratic automorphism

$$H_6(q, p) = (1 - 6q^2 - p, q),$$

we prove that every complex periodic point is real, hyperbolic, and simple. The proof combines three independent ingredients. A linear scaling places the map at parameter $(a, b) = (6, -1)$ in the standard Hénon family; Arai's certified hyperbolic plateau connects this parameter to a Devaney–Nitecki full-two-shift anchor; and the resulting 2^n distinct real fixed points of the nth iterate exhaust the Friedland–Milnor complex algebraic count 2^n. We then apply this exhaustion theorem to the odd mixed-axis closure polynomials introduced in two predecessor studies. Every such polynomial is totally real and squarefree, and its Möbius primitive quotient is a reduced effective divisor of exact least-period points. The primitive degree has entropy $\frac{1}{2}\log 2$. Exact Sturm computations certify all primitive roots through odd period 13, with an independent symbolic audit. The result resolves an ambient effectivity obstruction but does not supply rational-prime labels, a von Mangoldt trace, a completed Riemann determinant, or a Hilbert–Pólya operator.

1 Introduction

Periodic points of polynomial automorphisms carry two different kinds of structure. Their real dynamics may admit a symbolic coding and uniform hyperbolicity, while their complex coordinates form finite algebraic schemes. Ordinarily these descriptions need not exhaust one another: a real horseshoe can coexist with additional complex periodic points, and an algebraic intersection can contain multiplicity invisible to symbolic dynamics.

This paper identifies a parameter at which the two ledgers coincide exactly. Consider

$$H_6(q, p) = (1 - 6q^2 - p, q). \tag{1}$$

The map is a degree-two polynomial automorphism with Jacobian determinant one. Our main result is the following.

Theorem 1.1 (Algebraic exhaustion)**.** *For every integer $n \geq 1$, the complex fixed-point scheme of H_6^n is reduced and consists of exactly 2^n real uniformly hyperbolic points. Equivalently,*

$$\mathrm{Fix}_{\mathbb{C}}(H_6^n) = \mathrm{Fix}_{\mathbb{R}}(H_6^n), \qquad \#\,\mathrm{Fix}(H_6^n) = 2^n.$$

The proof is not a large finite computation. It is a short bridge among three source-native theorems. First, a linear conjugacy identifies (1) with the standard area-preserving Hénon map at $a = 6$. Arai's rigorous hyperbolic plateau contains the entire parameter path from 6 to 10 [1]. At the latter endpoint the Devaney–Nitecki large-parameter theorem gives a full two-shift [2]. Hyperbolic structural stability transports this symbolic system to $a = 6$, yielding 2^n distinct real fixed points. Second, Friedland–Milnor's intersection theorem says that the total complex algebraic multiplicity is also 2^n [3]. The real points therefore consume the entire complex count.

The motivation comes from a reflection slice developed in two predecessor projects. If $n = 2m+1$ is odd, an exact mixed-axis closure has the form

$$F_n(X) = q_{m+1}(X) - q_m(X), \qquad \deg F_n = 2^{m+1}, \tag{2}$$

where the q_j are generated by the Hénon recurrence on one reversor-fixed line. Formal Möbius division predicts a primitive degree of entropy $\frac{1}{2}\log 2$. Previous work established divisibility and transversality only on a smaller certified survivor. Theorem 1.1 removes the ambient gap.

Corollary 1.2 (Totally real primitive reflection divisors)**.** *For every odd n, F_n is totally real and squarefree. Its recursive Möbius quotient Ψ_n is a reduced effective divisor supported exactly on least-period-n points. In particular,*

$$\mathbb{Q}[X]/(\Psi_n)$$

is a finite étale *totally real algebra.*

The result is deliberately not phrased as an arithmetic spectral theorem. The total reality and exact entropy provide a stronger all-period object for future height and Galois-pressure questions, but they produce neither rational-prime labels nor the prime-power amplitudes required by a Riemann trace formula. The claim boundary is part of the theorem package.

Section 2 locks parameter conventions and source scopes. Section 3 transports the full horseshoe, and Section 4 proves complex algebraic exhaustion. Sections 5–6 derive reflection effectivity and entropy. Section 7 reports the exact audit, and Section 8 records the remaining arithmetic boundary.

2 Parameter and source locks

We use the standard area-preserving convention

$$H_{a,-1}(x, y) = (a - x^2 - y, x). \tag{3}$$

Define $S(q, p) = (6q, 6p)$. A direct calculation gives

$$S \circ H_6 \circ S^{-1}(x, y) = (6 - x^2 - y, x) = H_{6,-1}(x, y). \tag{4}$$

Thus every dynamical and algebraic statement can be transported through a real linear conjugacy.

Three imported results are needed. We state only the portions used here.

1. Arai proves that the chain recurrent set of $H_{a,-1}$ is uniformly hyperbolic for

$$a \in [5.699951171875, \infty) = \left[\frac{23347}{4096}, \infty\right), \tag{5}$$

and relates this hyperbolicity to R-stability on the plateau [1, Theorem 2].

2. Devaney and Nitecki use the normalization $(x, y) \mapsto (1 + y - Ax^2, Bx)$. At $B = -1$, the scaling $(X, Y) = (Ax, -Ay)$ gives $(X, Y) \mapsto (A - X^2 - Y, X)$, exactly (3). Their threshold $(5 + 2\sqrt{5})(1 + |B|)^2/4$ therefore reduces to

$$a > 5 + 2\sqrt{5} \tag{6}$$

[2].

3. Friedland and Milnor prove that a cyclically reduced polynomial automorphism of algebraic degree d has d fixed points over $\mathbb{C}$, counted with algebraic multiplicity [3, Theorem 3.1]. Applied to an iterate of a degree-two Hénon map, the count is 2^n.

The numerical work of Sterling, Dullin, and Meiss places the horseshoe boundary near 5.699 and supplies useful context [6], but it is not used as a theorem here. Likewise, general results on reversible periodic points and dynatomic cycles provide context [4, 5]; the mixed-axis identities used below are proved in the locked predecessor packages rather than imported from those papers.

The exact inequalities needed for the path are elementary:

$$6 > \frac{23347}{4096}, \qquad (10 - 5)^2 = 25 > 20 = (2\sqrt{5})^2.$$

Consequently $[6, 10]$ lies in Arai's connected hyperbolic region, and 10 lies strictly in the Devaney–Nitecki full-shift region.

3 Transporting the full horseshoe

Let $\mathcal{R}_a$ denote the real chain recurrent set of $H_{a,-1}$.

Theorem 3.1 (Full horseshoe at $a = 6$)**.** *The restriction $H_{6,-1}|_{\mathcal{R}_6}$ is uniformly hyperbolic and topologically conjugate to the full shift on two symbols.*

Proof. At $a = 10$, inequality (6) holds, so the Devaney–Nitecki theorem identifies the real invariant set with the full two-shift. By (5), every parameter in $[6, 10]$ has a uniformly hyperbolic chain recurrent set. Arai's R-stability statement provides conjugate chain recurrent dynamics along this connected path. Therefore the full-shift conjugacy at 10 continues to 6. Uniform hyperbolicity at 6 is also part of Arai's theorem. □

Transporting by (4) gives the same conclusion for H_6. The fixed points of the nth iterate of the full two-shift are precisely the 2^n binary words of length n with a marked origin. Hyperbolic conjugacy therefore gives the following count without any orbit-finding tolerance.

Corollary 3.2. *For every $n \geq 1$, H_6^n has exactly 2^n distinct real fixed points, and all of them are uniformly hyperbolic.*

Every real periodic point is chain recurrent, so no real periodic point lies outside the set covered by Theorem 3.1. The number of points of exact least period n and the number of primitive orbits are, respectively,

$$P_n = \sum_{d|n} \mu(n/d) 2^d, \qquad O_n = \frac{P_n}{n}. \tag{7}$$

4 Complex algebraic exhaustion

We now compare the real symbolic count with the complex intersection count.

Theorem 4.1. *For every $n \geq 1$, the complex fixed-point scheme of H_6^n is reduced and is supported on the 2^n real hyperbolic points from Corollary 3.2.*

Proof. The map H_6 is a cyclically reduced degree-two polynomial automorphism. The nth iterate has dynamical degree 2^n, and the Friedland–Milnor fixed-point theorem gives total complex algebraic multiplicity 2^n.

Corollary 3.2 supplies 2^n pairwise distinct real fixed points. Each contributes complex local intersection multiplicity at least one. Their total contribution is therefore at least 2^n, already equal to the complete Friedland–Milnor count. No other complex fixed point can exist, and every listed local multiplicity must equal one. Hyperbolicity was proved in Corollary 3.2. □

Remark 4.2. The comparison is multiplicity-sensitive. Equality of a topological entropy with $\log 2$ would not suffice. The proof uses an exact symbolic fixed-point count on one side and an exact complex intersection count on the other.

The same argument also identifies the exact-period scheme. Möbius inversion of the fixed-point counts gives (7); because every fixed-point scheme is reduced, this is an actual count of distinct exact-period points rather than a virtual degree.

Corollary 4.3. *No periodic point of H_6 has multiplier $+1$ or -1 on the unit circle. In particular, every complex periodic point is a real transverse hyperbolic point.*

Proof. Every complex periodic point is real by Theorem 4.1, and every real periodic point lies in the uniformly hyperbolic chain recurrent set. Uniform hyperbolicity excludes unit-modulus multipliers. □

5 Mixed-axis effectivity

We recall the exact reversible geometry. Let

$$R(q,p) = (p,q), \qquad J = RH_6.$$

Then $R^2 = J^2 = 1$ and $H_6 = RJ$. The fixed line of J has the parameterization

$$\gamma(X) = \left(X, \frac{1-6X^2}{2}\right). \tag{8}$$

For odd $n = 2m+1$, write

$$H_6^j\gamma(X) = (q_j(X), q_{j-1}(X)).$$

The mixed-axis closure is

$$F_n(X) = q_{m+1}(X) - q_m(X). \tag{9}$$

The predecessor divisibility theorem gives

$$\deg F_n = 2^{m+1} = 2^{(n+1)/2} \tag{10}$$

and shows that every complex root of F_n yields a complex point of $\mathrm{Fix}(H_6^n)$.

Theorem 5.1 (Total reality and squarefreeness)**.** *For every odd n, every root of F_n is real and simple.*

Proof. If $F_n(X) = 0$, the point $\gamma(X)$ is fixed by H_6^n. Theorem 4.1 makes this point real, hence its first coordinate X is real.

For simplicity, the predecessor transversality lemma applies to the two involution-fixed curves $\mathrm{Fix}(J)$ and $\mathrm{Fix}(H_6^{-(m+1)} R H_6^{m+1})$. A multiple closure root makes the two curves tangent. Their common tangent is fixed by both involution derivatives, hence by their product DH_6^n, forcing multiplier $+1$. Corollary 4.3 excludes this. Thus every root is simple. $\square$

Define the monic primitive quotients recursively over odd integers by

$$\Psi_1 = F_1, \qquad \Psi_n = F_n \Bigg/ \prod_{\substack{d \mid n \\ d < n}} \Psi_d. \tag{11}$$

The predecessor divisibility theorem proves that the quotient is a polynomial over $\mathbb{Q}$.

Theorem 5.2 (Exact primitive effectivity)**.** *For every odd n, Ψ_n is squarefree and its roots are exactly the mixed-axis points of least period n. Consequently $\mathbb{Q}[X]/(\Psi_n)$ is finite* étale *and totally real.*

Proof. Let X be a root of F_n and let e be the least period of $z = \gamma(X)$. Then $e \mid n$, and e is odd because n is odd. Since $z \in \mathrm{Fix}(J)$ and $H_6^e z = z$, reversibility gives

$$H_6^{(e+1)/2} z \in \mathrm{Fix}(R).$$

Thus X satisfies the mixed-axis closure $F_e(X) = 0$. Every lower-period root is therefore removed by one of the proper-divisor factors in (11).

Conversely, an exact-period-n mixed-axis root lies in F_n and in no F_d with proper $d \mid n$, so it remains in Ψ_n. Theorem 5.1 makes the complete family of closure roots real and simple. Hence the quotient is reduced, its support is exactly the claimed set, and the quotient algebra is finite *étale* and totally real. $\square$

6 Primitive reflection entropy

Taking degrees in (11) and applying Möbius inversion gives an actual root count, not merely a formal divisor degree:

$$D_n := \deg \Psi_n = \sum_{d \mid n} \mu(n/d) 2^{(d+1)/2}. \tag{12}$$

Proposition 6.1. *Along odd integers,*

$$D_n = 2^{(n+1)/2} + O\left(n 2^{n/6+1/2}\right), \qquad \lim_{\substack{n\to\infty \\ n \,\mathrm{odd}}} \frac{1}{n} \log D_n = \frac{1}{2} \log 2.$$

Proof. The $d = n$ term in (12) is $2^{(n+1)/2}$. Every proper divisor of an odd n is at most $n/3$. There are at most n divisors, so the absolute value of the remaining sum is bounded by $n2^{n/6+1/2}$. The entropy limit follows. $\square$

The earlier small-content survivor contains a distinguished physical reflection population with count $P_n = \Theta(\varphi^{n/2})$. Theorem 5.2 changes the interpretation of the ratio

$$\frac{P_n}{D_n} = \Theta\Big((\varphi/2)^{n/2}\Big). \tag{13}$$

Previously the denominator was only a formal degree. It is now the count of a reduced all-real hyperbolic population. The physical population is therefore genuinely exponentially sparse inside the ambient reflection slice.

This does not make the ambient population arithmetic. It only replaces an effectivity obstruction by a clean height-pressure question on a certified sequence of totally real finite *étale* algebras.

7 Finite exact audit

The executable certificate is an audit of the theorem chain, not its source. It reconstructs (9) over $\mathbb{Q}[X]$, performs the exact division (11), and applies Sturm isolation to every primitive root through odd period 13.

Table 1: Exact primitive reflection audit.

n	deg F_n	deg Ψ_n	real simple roots	exact half-words
1	2	2	2	2
3	4	2	2	2
5	8	6	6	6
7	16	14	14	14
9	32	28	28	28
11	64	62	62	62
13	128	126	126	—

For $n \leq 11$, every rational isolating interval is propagated through the Hénon recurrence by interval arithmetic, producing a unique sign half-itinerary. Period 13 is intentionally marked theorem-independent for that optional diagnostic: exact rational propagation produces very large denominators, while Sturm theory already certifies all 126 roots without it.

An independent implementation enumerates all binary full-shift words through period 13 and separately computes high-precision roots through period 11. The primary checker also freezes the predecessor artifacts by SHA-256 and rejects 26 mutations of the parameter bridge, source scopes, algebraic count, effectivity, entropy, Route-A status, and claim boundary. Ten unit tests cover recurrence, division, root counts, dependencies, and mutation execution.

All exact outputs are serialized in canonical JSON. No prime table, Riemann zero table, fitted parameter, random seed, or GPU computation enters the certificate.

8 Route implications and arithmetic boundary

Theorem 4.1 is a genuine all-period Route-A advance. It supplies a complete primitive periodic-point layer, an exact symbolic zeta on the real chain recurrent set, and totally real reflection algebras. In the repository evaluation vocabulary, the strongest tuple is

$$(A1_{\text{analytic}}, A2_{\text{inherited symbolic}}, A3_{\text{partial}}, A4_{\text{formal}}).$$

The overall verdict remains `ROUTE_A_EXPLORATORY`.

The following statements are *not* consequences of the theorem:

1. there is no intrinsic bijection between primitive Hénon orbits and rational primes;
2. no amplitude $\log p/p^{r/2}$ or von Mangoldt trace has been derived;
3. the full-shift zeta is not the completed Riemann ξ function;
4. total reality of finite algebras is not self-adjointness of one infinite-dimensional operator;
5. no Hilbert space, operator domain, compact resolvent, or zeta-regularized determinant identity has been constructed.

Route B is therefore not testable and is not authorized.

The next non-micro object is an all-period algebraic pressure. If $h(\alpha)$ is a canonical height or Galois-excess statistic on roots of Ψ_n, one may ask for the existence and regularity of

$$\mathcal{P}_{\mathrm{Gal}}(s) = \limsup_{\substack{n\to\infty \\ n \text{ odd}}} \frac{1}{n} \log \sum_{\Psi_n(\alpha)=0} e^{-sh(\alpha)}.$$

A comparison with the physical instability pressure would be meaningful without forcing a termwise prime labeling. This is the narrow successor problem retained by the present paper.

9 Conclusion

At the area-preserving parameter H_6, a real full horseshoe and a complex algebraic intersection count agree exactly at every iterate. Their equality forces every complex periodic point to be real, hyperbolic, and simple. The result converts the odd mixed-axis closure sequence from a formal divisor into a reduced effective sequence of totally real primitive algebras with entropy $\frac{1}{2}\log 2$.

This closes the ambient effectivity problem isolated by the preceding two papers. It also sharpens the next obstruction: the missing bridge is no longer real-root existence or transversality, but a uniform arithmetic height-pressure law and, beyond that, a source-native trace. Those steps are left open rather than inferred from total reality.

References

[1] Zin Arai. On hyperbolic plateaus of the hénon map. *Experimental Mathematics*, 16(2):181–188, 2007. doi: 10.1080/10586458.2007.10128992. URL https://www.math.kyoto-u.ac.jp/preprint/2005/8arai.pdf.

[2] Robert L. Devaney and Zbigniew Nitecki. Shift automorphisms in the hénon mapping. *Communications in Mathematical Physics*, 67:137–146, 1979. doi: 10.1007/BF01221362.

[3] Shmuel Friedland and John Milnor. Dynamical properties of plane polynomial automorphisms. *Ergodic Theory and Dynamical Systems*, 9(1):67–99, 1989. doi: 10.1017/S014338570000482X.

[4] Benjamin Hutz. Effectivity of dynatomic cycles for morphisms of projective varieties using deformation theory, 2010.

[5] Jungsoo Kang. On reversible maps and symmetric periodic points, 2014.

[6] David Sterling, Holger R. Dullin, and James D. Meiss. Homoclinic bifurcations for the hénon map. *Physica D*, 134(2):153–184, 1999. doi: 10.1016/S0167-2789(99)00125-6.

A Proof and claim ledger

Claim	Status	Basis
Parameter conjugacy	proved exact	direct symbolic calculation
Hyperbolic plateau	source-backed proved	Arai, Theorem 1.2
Full-shift anchor	source-backed proved	Devaney–Nitecki at $a = 10$
Full shift at $a = 6$	proved from sources	connected R-stable plateau
Complex count 2^n	source-backed proved	Friedland–Milnor, Theorem 3.1
Algebraic exhaustion	proved	equality of distinct real count and complex multiplicity
Closure total reality	proved	every closure root is a periodic point
Closure simplicity	proved	reversor tangency plus hyperbolicity
Primitive effectivity	proved	exact divisibility and least-period removal
Periods through 13	computer-certified exact	rational algebra and Sturm isolation
Prime trace / ξ identity	open	no source-native compiler
Hilbert–Pólya operator	not testable	no operator/domain

The source-backed rows are inputs; the exhaustion and reflection rows are the new deductions of this paper. The finite certificate audits but does not replace the all-period proof.